CREWFORGE

LEVEL 1 · MW-001 · RED SEAL INDUSTRIAL MECHANIC (MILLWRIGHT)

Sling Angles and Load Angle Factors

Why a sling rated for the load can still be overloaded

Learner _____

CrewForge textbook chapter · Rigging, Hoisting, Lifting and Moving (IMM-150) · Free edition · CrewForge Rigging Starter

Exam preparation, not a procedure. Always follow the sling tag, the manufacturer's data and your lift plan.

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About this chapter

This chapter covers one of the most tested ideas in rigging: how the angle of a sling changes the load it carries. You'll learn why it happens, how to calculate it, how to measure it on the job, and how it changes with every common hitch. Work through the examples with a calculator. Then try the practice problems at the end without looking at the answers.

Every number in this chapter uses one method: the load angle factor rounded to three decimals, and loads rounded to the nearest pound or kilogram.

The question that catches people. You've got a 10,000 lb load and two slings, each rated 8,000 lb. That's 16,000 lb of sling for a 10,000 lb load. It sounds like plenty. Rig those slings at 30° and each leg carries the full 10,000 lb: both slings are now 25 % overloaded. The rating didn't change. The angle did.

What you'll learn

- Explain what a sling's working load limit means, and why the design factor is not spare capacity.
- Explain, using force parts, why lowering the sling angle increases the tension in each leg.
- Calculate the load angle factor for any angle, and the tension in each leg, in imperial and metric.
- Convert an angle given from vertical, or an included angle, into the angle from horizontal.

- Find the load angle factor in the field with a tape measure (L ÷ H), and find H from the sling length and the distance between pick points.
- Work backwards from sling ratings to the most a rig can lift.
- Apply choker, basket and multi-leg bridle rules, including the angle-of-choke reduction and the two-leg rule.
- Find the load in each leg when the centre of gravity is off centre.
- Plan a complete two-leg lift, and name every part of the load path that must be rated for the force it sees.

Key terms

Working load limit (WLL): The most a sling or fitting is rated to carry in a stated hitch or pull, from its tag or stamp. Also called the rated load or rated capacity.

WLL: Short for working load limit.

Rated capacity: Another name for the WLL: the most the sling may carry in the stated hitch.

Hitch: The way a sling is put on a load and the hook: vertical, choker, basket or bridle.

Vertical hitch: One sling straight up and down from the hook to one pick point. Tags give this rating first.

Load path: Every part the load hangs from, from the hook down to the load: hook, master link, slings, shackles and lugs.

Signal person: The one worker who gives the crane or hoist operator signals during the lift.

Design factor: Breaking strength divided by the rated load. It covers wear, shock loading and small errors. It is not extra capacity for you to use.

Leg: One part of a sling, or one sling, running from the hook to a pick point.

Pick point: A place on the load where a sling is attached: a lifting lug, eye bolt or a point the sling passes around.

Master link: The large link at the top of a bridle that sits in the hook and holds all the legs.

Sling angle (θ): The angle between the sling leg and the horizontal. Every formula in this chapter uses this angle.

Angle from vertical: The angle between the sling leg and a vertical line. $\theta = 90° - \text{angle from vertical}$.

Included angle: The full angle between two legs at the hook. $\theta = 90° - (\text{included angle} \div 2)$.

Load angle factor (LAF): The multiplier that the sling angle puts on leg tension: $LAF = 1 \div \sin\theta = L \div H$.

Tension (T): The pull along a sling leg.

Sine (sin): For an angle in a right-angled triangle: the upright side divided by the long side. For a sling leg, height ÷ length.

Tangent (tan): For an angle in a right-angled triangle: the upright side divided by the flat side. For a sling leg, height ÷ sideways distance.

L and H: L is the sling leg length. H is the height from the pick points up to the hook (the master link).

Bridle: Two, three or four legs from one master link to separate pick points.

Choker hitch: The sling passes around the load and through its own eye, closing on the load as it lifts.

Angle of choke: The angle the sling makes where it passes through its own eye in a choker hitch.

Basket hitch: The sling supports the load from underneath, with both eyes on the hook.

D/d ratio: The diameter the sling bends around (D) divided by the sling's own diameter (d). A small D/d lowers the sling's capacity.

Two-leg rule: Plan a three- or four-leg bridle as if only two legs carry the whole load.

Centre of gravity (CG): The point where the whole weight of the load acts. The hook must be directly above it for a level lift.

Spreader beam: A beam below the hook that holds the slings apart so they hang closer to vertical.

1 What the sling rating means

Every calculation in this chapter ends with one question: is each leg within its rating?

Every sling has a tag. The tag gives the maker, the sling's size and material, and its **working load limit (WLL)** for the hitches it is rated for. Bridle tags often also list the rating at 60°, 45° and 30°. If the tag is missing or can't be read, the sling is taken out of service until it is re-tagged by the maker or a qualified person.

The WLL is far below the force that would break the sling. The ratio between them is the **design factor**. Under ASME B30.9, alloy chain slings use a design factor of 4, and wire rope and synthetic slings use 5. A 5,000 lb wire rope sling would break somewhere near 25,000 lb when new.

That margin is not yours to use. It covers things you can't see or control: wear, small cuts, heat, shock loading when a load is jerked, and small errors in weight and angle. Rig to the WLL, every time.

SLING MATERIAL	DESIGN FACTOR (ASME B30.9)	CHOKER RATING, % OF VERTICAL (TYPICAL)
Wire rope	5 : 1	75 %
Alloy chain (grade 80 / 100)	4 : 1	80 %
Synthetic web and roundsling	5 : 1	80 %

Choker figures are typical values used in maker catalogues and in the CrewForge simulator. Always use the sling tag.

> SAFETY
> **No tag, no lift**
> A sling with a missing or unreadable tag is not used. Inspect every sling before each lift: look for cuts, broken wires, kinks, stretched or nicked links, heat damage and damaged fittings.

Each of these faults means the sling has lost strength you can't measure on the job, so the rating on the tag no longer holds. A damaged sling can fail well below its WLL. OSHA's sling guidance says a qualified person checks slings each day before use, and takes a sling out of service at once for any of these:

SLING	TAKE IT OUT OF SERVICE FOR	HOW YOU CHECK
Wire rope	Broken wires; kinks, crushing or bird caging; heat damage; crushed or worn end fittings; severe corrosion	Look along the whole length, the eyes and the fittings
Alloy chain	Nicks, cracks, gouges, stretch or bends in any link; heat discolouration; pitting or corrosion	Look at each link and the hook
Synthetic web or roundsling	Cuts, holes, tears or snags; melting, charring or acid burns; broken stitching; knots; stiff or discoloured areas	Look along both faces and the eyes
Every sling	A missing or unreadable tag	Read the tag

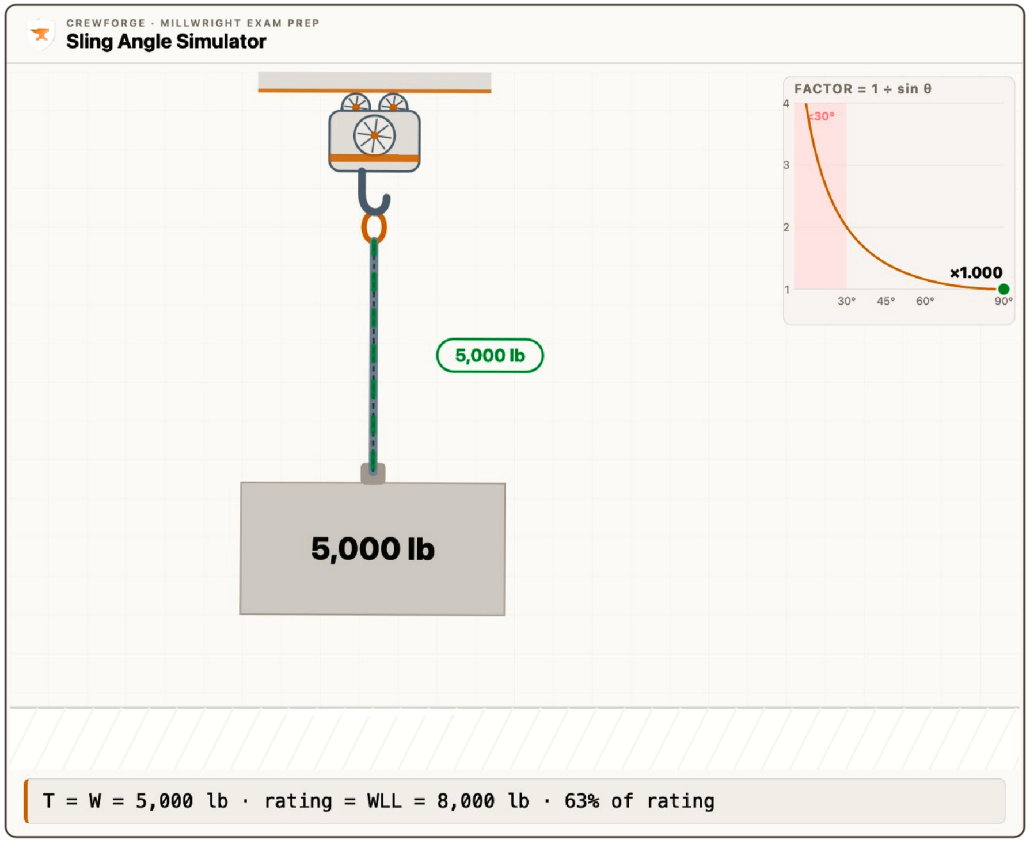
OSHA, Guidance on Safe Sling Use (wire rope, alloy chain and synthetic web pages): removal from service.

The rest of the chapter answers the question the tag can't: how much tension will this leg actually see in the rig you're about to build?

2 Why the angle adds tension

Most rigging overloads don't come from heavy loads. They come from low sling angles.

A sling can only pull along its own length. When two slings hang straight down, all of each leg's pull holds the load up, and each leg carries exactly half the load.

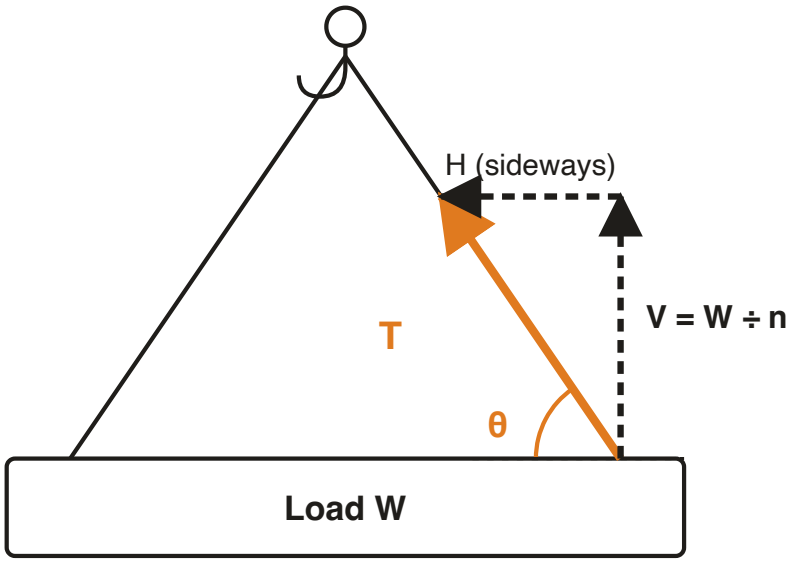


Vertical: the leg carries exactly the load. 5,000 lb load, 5,000 lb in the sling.

Now move the pick points apart, or shorten the slings, so the legs lean in toward the hook. Each leg still pulls along its own length, but now that pull points partly up and partly sideways.

Splitting the pull into two parts

Think of the tension in a leg as two parts working together. The **vertical part** holds the load up. The **horizontal part** pulls the pick point sideways, toward the other leg. The two parts and the tension form a right-angled triangle.



Force parts in one leg of a two-leg bridle. The vertical part is always the leg's share of the load (W ÷ n). The tension T is the long side of the triangle.

The vertical part never changes. However the sling is rigged, the legs together must hold up the whole load, so each leg's vertical part is the load divided by the number of legs: **V = W ÷ n**.

Two words from the triangle do the work. The **sine** of the angle (sin θ) is the upright side divided by the long side. The **tangent** (tan θ) is the upright side divided by the flat side. A calculator gives both: Part 3 shows the keys.

The lower the sling angle, the more of the leg's pull goes sideways. To keep the same vertical part, the tension has to rise. In the triangle, the vertical part is T × sin θ. So:

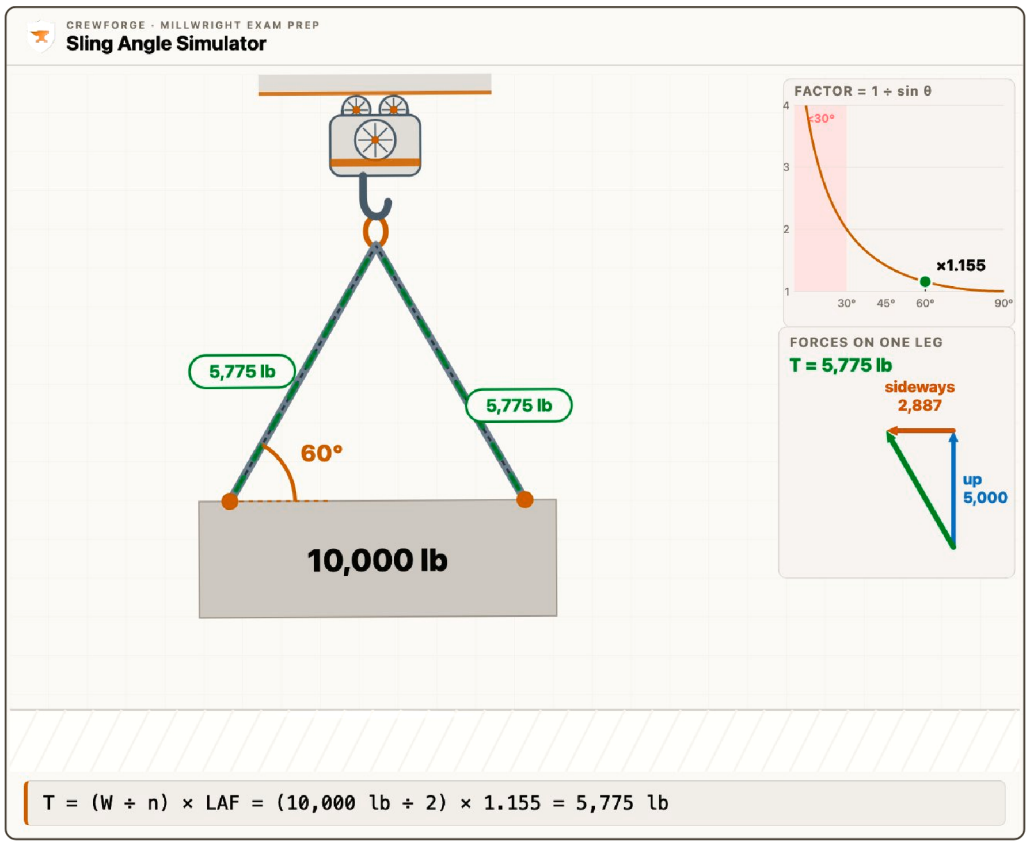
> **V = T × sin θ**
> **T = V ÷ sin θ = (W ÷ n) × (1 ÷ sin θ)**

EXAMPLE 2.1 **Tension, split into its upward and sideways parts**

A 6,000 lb load hangs from a two-leg bridle. Each leg is at 60° from horizontal. Find the tension in each leg and the horizontal part of that tension.

1. Vertical part of each leg: 6,000 ÷ 2 = 3,000 lb
2. Load angle factor at 60°: 1 ÷ sin 60° = 1 ÷ 0.866 = 1.155
3. Tension: 3,000 × 1.155 = 3,465 lb
4. Horizontal part: V ÷ tan θ = 3,000 ÷ tan 60° = 3,000 ÷ 1.732 = 1,732 lb

Answer: 3,465 lb in each leg; each pick point is pulled about 1,732 lb toward the other.



At 60°, each leg's pull splits into an upward part (always half the load) and a sideways part.

ON THE JOB

The sideways pull matters too

The horizontal part pulls the pick points toward each other. On a long, slim load such as a pipe spool, a beam or a light frame, that inward pull can bend the load. Lifting lugs and eye bolts made for a straight pull can be overloaded or bent when they're pulled at an angle. Check the lug or eye bolt rating for the angle you're using, or use a spreader beam.

CHECK YOUR UNDERSTANDING

1. Two slings hang straight down from one hook to a 4,000 lb load. What is the tension in each?
2. As the sling angle gets lower, what happens to the vertical part of each leg's pull?

Answers at the back of the chapter.

3 The load angle factor

One formula answers almost every sling angle question on the exam.

The part of the formula that depends only on the angle is called the **load angle factor (LAF)**. It tells you how many times larger the leg tension is than the leg's share of the load.

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LAF = 1 ÷ sin θ
T = (W ÷ n) × LAF
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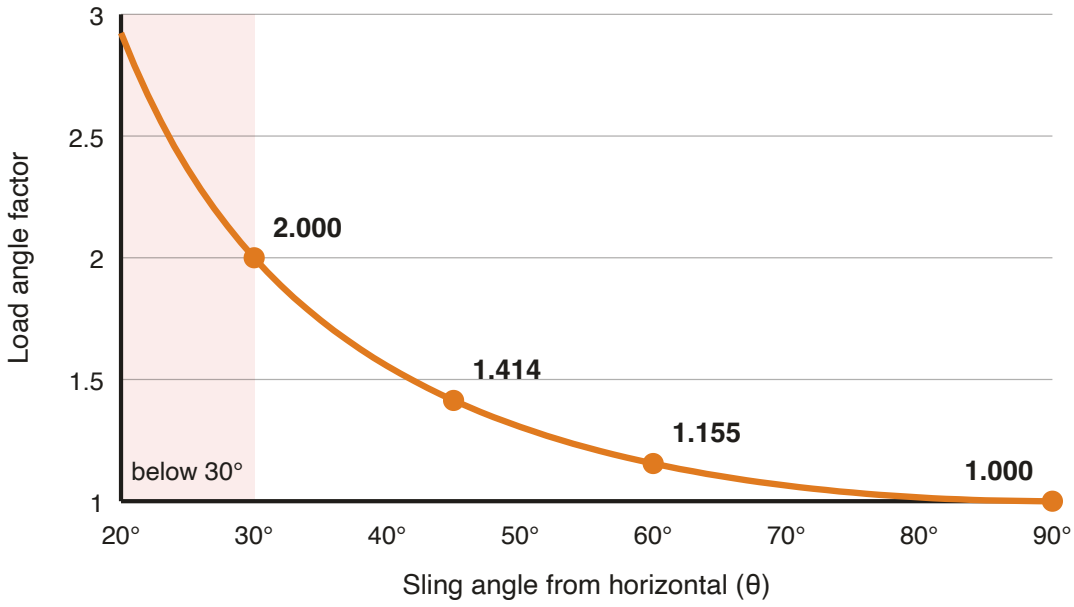
Here θ is the sling angle measured from horizontal, W is the load and n is the number of legs that carry it. The factor is always 1 or more. At 90° (vertical) it is exactly 1.

What sin means, and how to get it

Sin is short for sine. For a sling leg, sin θ is the height the leg climbs divided by its length. A leg at 60° climbs 0.866 ft for every foot of sling, so sin 60° = 0.866. The steeper the leg, the closer sin θ gets to 1. On a calculator set to degrees (DEG), press sin then 60, or 60 then sin on some models, and read 0.866. Then 1 ÷ 0.866 = 1.155, the load angle factor.

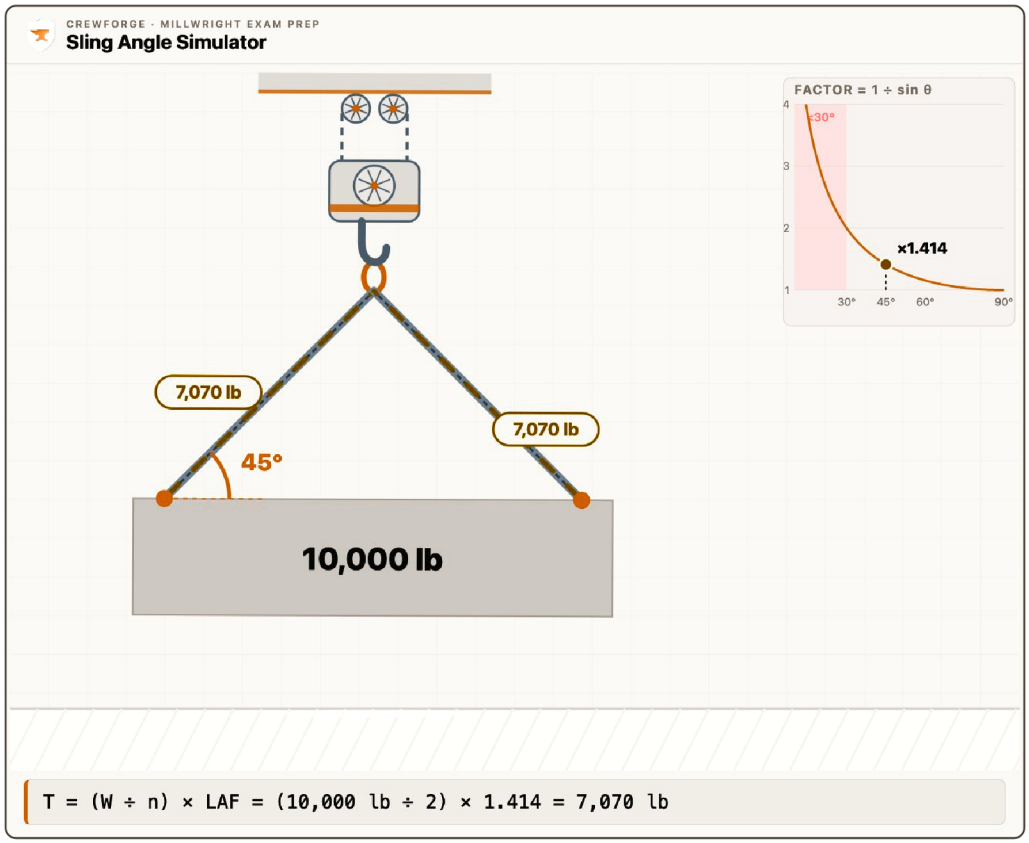
ANGLE FROM HORIZONTAL	LOAD ANGLE FACTOR	TENSION PER LEG, 10,000 LB ON TWO LEGS	INCREASE OVER VERTICAL
90°	1.000	5,000 lb	0 %
85°	1.004	5,020 lb	0.4 %
80°	1.015	5,075 lb	1.5 %
75°	1.035	5,175 lb	3.5 %
70°	1.064	5,320 lb	6.4 %
65°	1.103	5,515 lb	10.3 %
60°	1.155	5,775 lb	15.5 %
55°	1.221	6,105 lb	22.1 %
50°	1.305	6,525 lb	30.5 %
45°	1.414	7,070 lb	41.4 %
40°	1.556	7,780 lb	55.6 %
35°	1.743	8,715 lb	74.3 %
30°	2.000	10,000 lb	100 %

Factors are 1 ÷ sin θ rounded to three decimals. Use the same rounding in every step.

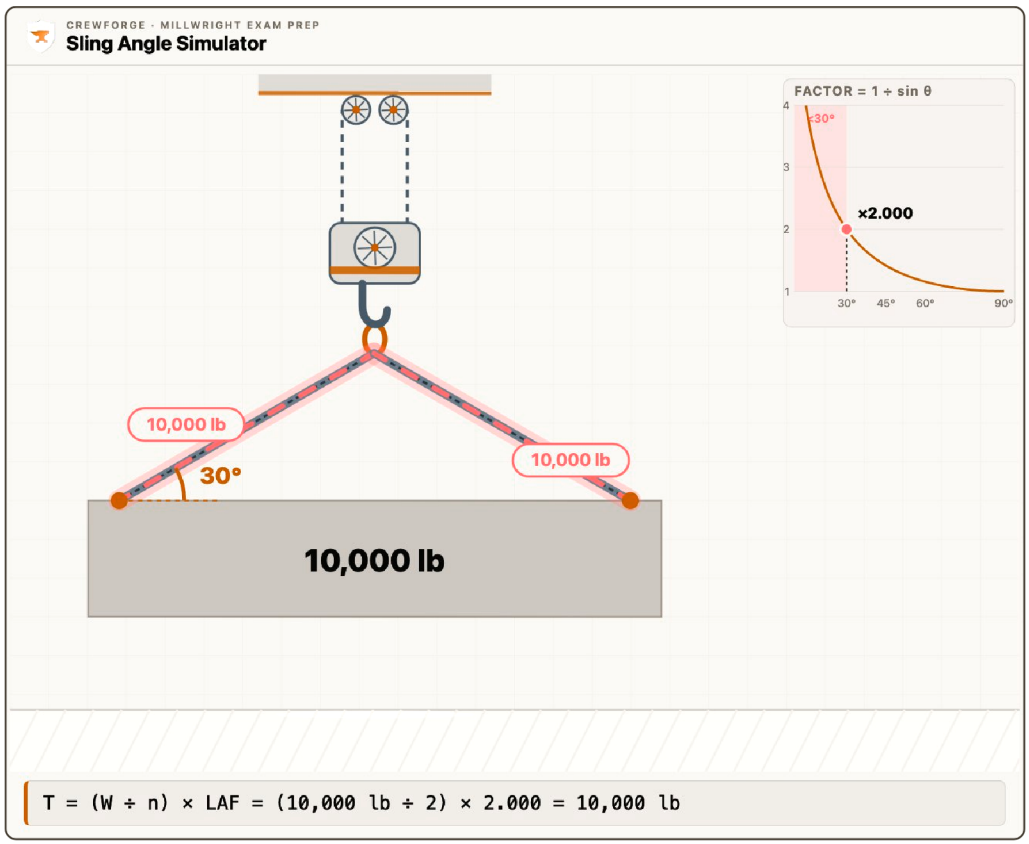


The load angle factor climbs slowly at first, then steeply below about 45°. Below 30° the climb is too steep to rig without an engineered lift plan.

Four factors are worth memorising: **90° = 1.000, 60° = 1.155, 45° = 1.414, 30° = 2.000**. At 60° each leg carries about 15 % more than its share. At 45°, about 41 % more. At 30°, double.



45°: factor 1.414. Each leg now carries 7,070 lb, over 80 % of its 8,000 lb rating.



30°: factor 2.000. Each leg carries the full 10,000 lb. Two 8,000 lb slings are overloaded.

EXAMPLE 3.1 **Imperial, angle in the table**

An 8,000 lb motor is lifted with a two-leg bridle at 45° from horizontal. What is the tension in each leg?

1. Share per leg: 8,000 ÷ 2 = 4,000 lb
2. 45° → LAF 1.414
3. 4,000 × 1.414 = 5,656 lb

Answer: 5,656 lb per leg

EXAMPLE 3.2 **Metric**

A 3,200 kg pump is lifted with a two-leg bridle at 45°. What is the tension in each leg, in kilograms and in kilonewtons?

1. Share per leg: 3,200 ÷ 2 = 1,600 kg
2. 45° → LAF 1.414
3. 1,600 × 1.414 = 2,262 kg
4. In kilonewtons: 2,262 × 9.81 ÷ 1,000 = 22.2 kN

Answer: 2,262 kg (22.2 kN) per leg

EXAMPLE 3.3 **An angle that isn't in the table**

A 12,000 lb heat exchanger bundle is lifted on two legs at 50° from horizontal. Find the tension in each leg.

1. Share per leg: 12,000 ÷ 2 = 6,000 lb
2. LAF = 1 ÷ sin 50° = 1 ÷ 0.7660 = 1.305
3. 6,000 × 1.305 = 7,830 lb

Answer: 7,830 lb per leg

METRIC

Kilograms, tonnes and kilonewtons

Metric sling tags give the rating in kilograms or tonnes (1 t = 1,000 kg). Work in the same unit as the tag. If an engineer gives a force in kilonewtons, convert with kN = kg × 9.81 ÷ 1,000. One tonne is about 9.81 kN.

EXAM TIP

Exam tip: degrees mode

Check that your calculator is in degrees, not radians. In radians, sin 60 gives –0.305 and every answer after that is wrong. A quick test: sin 30 must give 0.5.

SAFETY

Below 30°

Below 30° from horizontal the tension rises faster than most rigs can handle, and small angle errors make big tension errors. Don't rig below 30° without a qualified person's approval and an engineered lift plan.

CHECK YOUR UNDERSTANDING

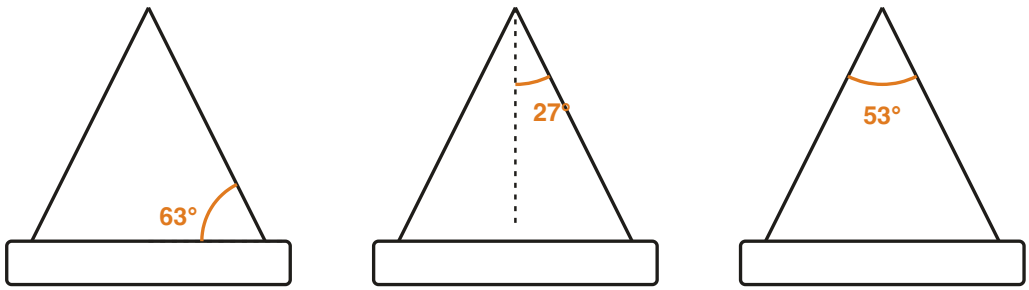
1. Find the load angle factor at 70°.
2. A 5,000 lb load hangs on two legs at 60°. What is the tension in each leg?
3. A 1,800 kg load hangs on two legs at 30°. What is the tension in each leg?

Answers at the back of the chapter.

4 Which angle did they give you?

Exam questions often give the angle from vertical, or the included angle, to see if you convert it.

There are three common ways to describe the same sling angle. The formulas in this chapter all need the **angle from horizontal**, so convert first.

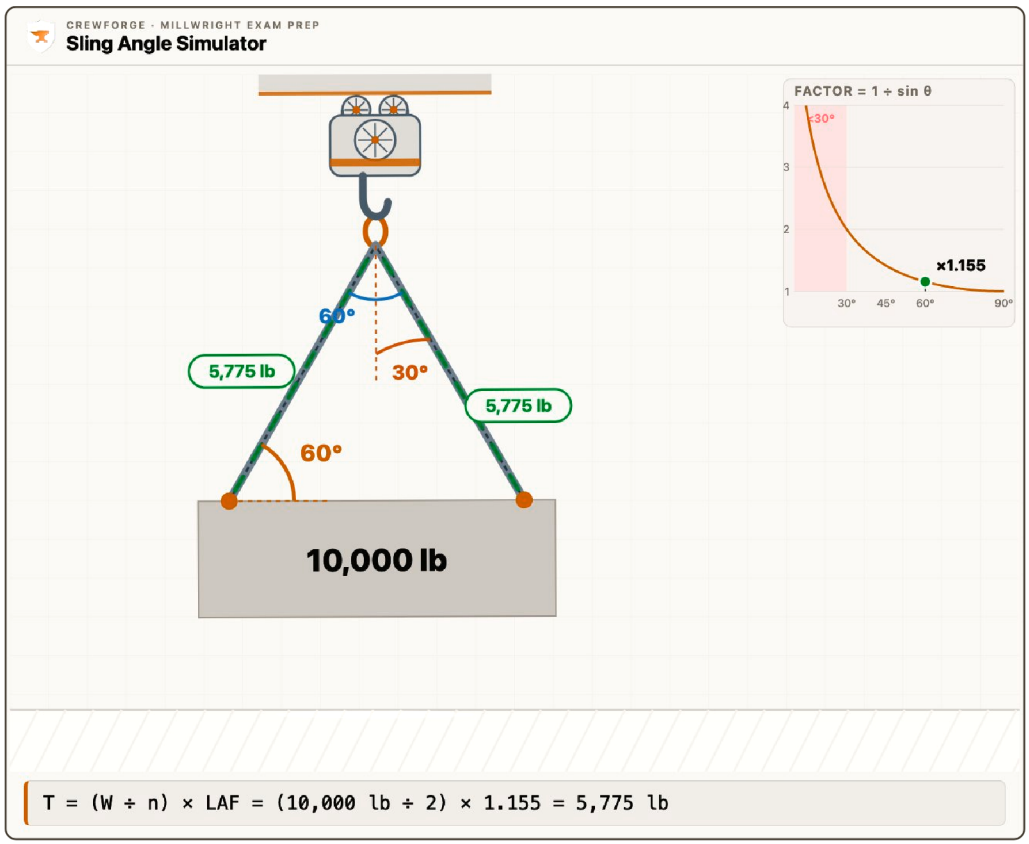


The same rig described three ways. Only the angle from horizontal goes into the formula.

```
θ = 90° − angle from vertical
θ = 90° − (included angle ÷ 2)
```

ANGLE FROM HORIZONTAL	ANGLE FROM VERTICAL	INCLUDED ANGLE	LAF
90°	0°	0°	1.000
60°	30°	60°	1.155
45°	45°	90°	1.414
30°	60°	120°	2.000

Useful pairs to recognise on sight.



60° from horizontal is 30° from vertical, and a 60° included angle.

EXAMPLE 4.1 **Angle from vertical**

An 8,000 lb load is lifted on two legs. Each leg is 25° from vertical. Find the tension in each leg.

1. Convert: θ = 90° − 25° = 65° from horizontal
2. LAF = 1 ÷ sin 65° = 1.103
3. (8,000 ÷ 2) × 1.103 = 4,412 lb

Answer: 4,412 lb per leg

EXAMPLE 4.2 **Included angle, metric**

A 5,000 kg load is lifted on two legs with a 90° included angle at the hook. Find the tension in each leg.

1. Convert: θ = 90° − (90° ÷ 2) = 45°
2. 45° → LAF 1.414
3. (5,000 ÷ 2) × 1.414 = 3,535 kg

Answer: 3,535 kg per leg

EXAMPLE 4.3 **The trap answer**

A question gives a 10,000 lb load on two legs, each 30° from vertical. A learner uses 30° in the formula. What answer do they get, and what is the right answer?

1. Wrong: (10,000 ÷ 2) × LAF(30°) = 5,000 × 2.000 = 10,000 lb
2. Right: θ = 90° − 30° = 60°, so 5,000 × 1.155 = 5,775 lb

Answer: The right answer is 5,775 lb. The trap answer, 10,000 lb, is almost always one of the choices.

EXAM TIP

Exam tip: read the angle twice

Before you calculate, underline the angle and ask: from horizontal, from vertical, or between the legs? The wrong angle gives a believable answer, and it will be on the answer sheet.

CHECK YOUR UNDERSTANDING

1. A leg is 40° from vertical. What is the sling angle from horizontal?
2. The included angle is 120°. What is the sling angle, and the load angle factor?

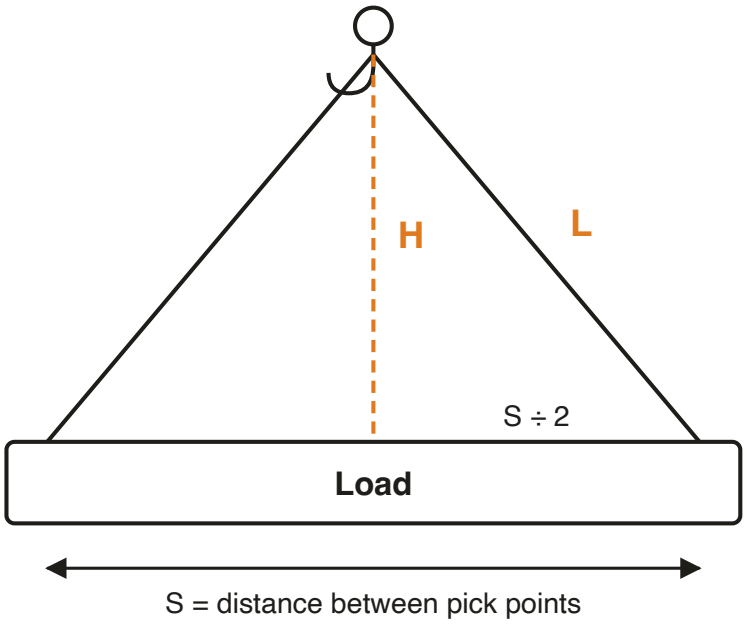
Answers at the back of the chapter.

5 Measuring the angle on the job

In the field you rarely have a protractor, but you always have a tape.

In the right-angled triangle formed by a sling leg, the sling is the long side (L) and the height from the pick point up to the hook is the upright side (H). So sin θ = H ÷ L, and the load angle factor comes straight from two measurements:

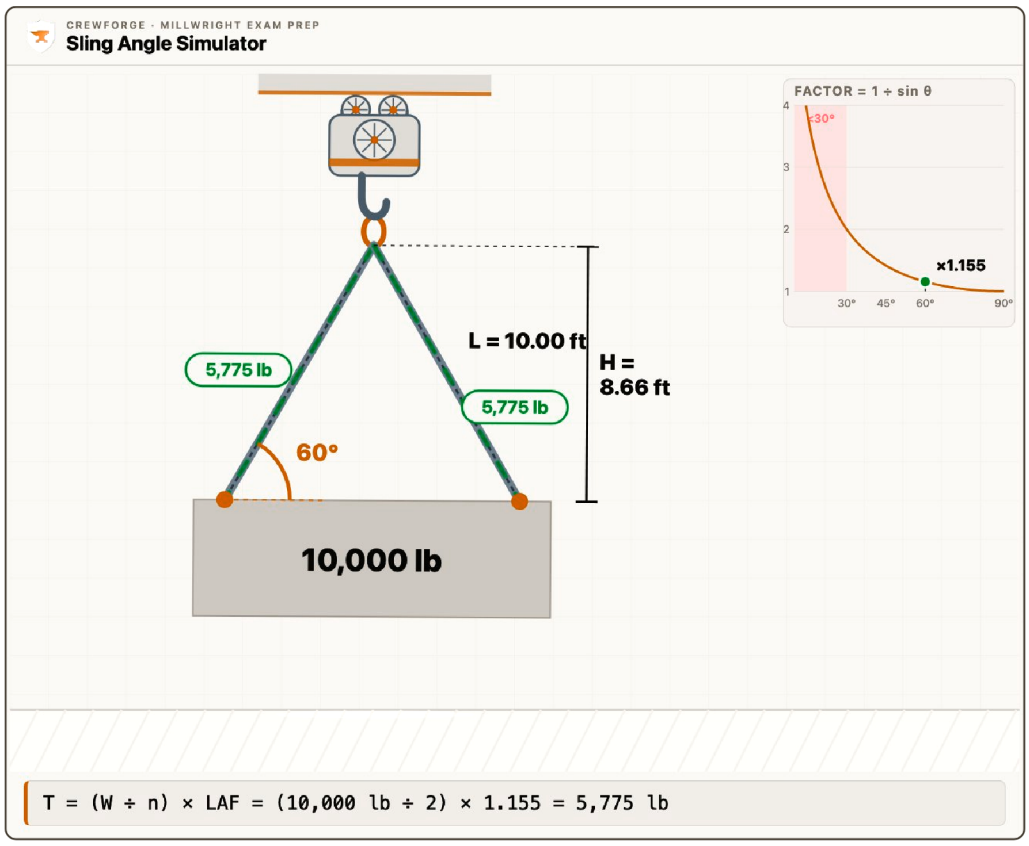
```
sin θ = H ÷ L
LAF = 1 ÷ sin θ = L ÷ H
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Measure L along the sling, and H straight up from the pick points to the hook. For a symmetric two-leg bridle, H can also be found from L and the distance S between pick points.

If you can't measure H directly, find it from the sling length and the distance between the pick points (S). For a symmetric two-leg bridle, each leg covers half of S sideways. Pythagoras says that in a right-angled triangle the long side squared equals the other two sides squared, added: $L^2 = H^2 + (S \div 2)^2$. Turn that round to get H. On a calculator, square each number, subtract, then press the square root key (√):

```
H = √(L² − (S ÷ 2)²)
```



L = 10.00 ft and H = 8.66 ft: L ÷ H = 1.155, the same factor as 60°.

EXAMPLE 5.1 **Using the pick-point distance**

The drawing shows pick points 10 ft apart. Two 10 ft slings lift a 7,200 lb load from them. Find the load angle factor and the tension in each leg.

1. H = √(10² − 5²) = √75 = 8.660 ft
2. LAF = L ÷ H = 10 ÷ 8.660 = 1.155 (that's 60°)
3. (7,200 ÷ 2) × 1.155 = 4,158 lb

Answer: LAF 1.155; 4,158 lb per leg

EXAMPLE 5.2 **Metric, an angle you didn't plan**

The drawing shows lugs 4 m apart. Two 3 m slings lift a 2,000 kg load from them. Find H, the load angle factor, the sling angle and the tension in each leg.

1. H = √(3² − 2²) = √5 = 2.236 m
2. LAF = 3 ÷ 2.236 = 1.342
3. Sling angle: sin θ = 2.236 ÷ 3, so θ = 48.2°
4. (2,000 ÷ 2) × 1.342 = 1,342 kg

Answer: H = 2.24 m; LAF 1.342; about 48°; 1,342 kg per leg

Headroom and shorter slings

When there isn't enough headroom, it's tempting to use shorter slings. Shorter slings on the same pick points make the angle lower, and the tension goes up.

EXAMPLE 5.3 **Shorter slings, same pick points**

The 7,200 lb load above won't clear an obstruction with 10 ft slings, so someone swaps in 7 ft slings on the same pick points, 10 ft apart on the drawing. Find the new tension in each leg.

1. $H = \sqrt{(7^2 - 5^2)} = \sqrt{24} = 4.899$ ft
2. LAF = 7 ÷ 4.899 = 1.429 (about 44°)
3. (7,200 ÷ 2) × 1.429 = 5,144 lb
4. Compare: 4,158 lb with 10 ft slings

Answer: 5,144 lb per leg, 24 % more than with the longer slings

ON THE JOB

Short on headroom?

The answer is a spreader beam, a different pick, or a different lift plan, not shorter slings. A spreader beam holds the slings below it close to vertical, so the factor on those legs stays near 1.

EXAM TIP

Exam tip: L over H, never H over L

The factor is always 1 or more. If L ÷ H gives you less than 1, you've divided the wrong way.

CHECK YOUR UNDERSTANDING

1. A sling is 8 ft long and the hook is 6.4 ft above the pick points. What is the factor?
2. Two 4 m slings; the drawing shows the lugs 2 m apart. What is H?

Answers at the back of the chapter.

6 Is the rig strong enough?

On the job the question is usually not 'what is the tension?' but 'can these slings make this lift?'

A rig is within its rating when the tension in every leg is no more than that leg's WLL for the hitch used. You can check it two ways: work out the tension and compare it with the WLL, or work backwards from the WLL to the most the rig can lift.

> **Each leg: T ≤ WLL**
> **Most a two-leg bridle can lift = 2 × WLL ÷ LAF**

EXAMPLE 6.1 **Working backwards from the rating**

Two wire rope slings, each rated 5,000 lb vertical, are rigged as a bridle at 45°. What is the most they can lift? Can they lift a 7,500 lb load?

1. Most the rig can lift: 2 × 5,000 ÷ 1.414 = 7,072 lb
2. 7,500 lb is more than 7,072 lb

Answer: 7,072 lb. No: the 7,500 lb load would overload both legs at 45°.

EXAMPLE 6.2 **Picking slings from the crib**

The crib has wire rope slings rated 3,200 lb, 4,800 lb and 6,400 lb. You need a two-leg bridle at 60° for a 9,200 lb load. Which slings do you use?

1. Share per leg: 9,200 ÷ 2 = 4,600 lb
2. 4,600 × 1.155 = 5,313 lb
3. 4,800 lb is below 5,313 lb, so it's not enough
4. 6,400 lb is above it

Answer: The 6,400 lb slings: each leg sees 5,313 lb, 83 % of its rating.

> ON THE JOB
> **The whole load path**
> The slings aren't the only part in the load path. The hook, the master link, every shackle and every lug or eye bolt must also be rated for the force it sees. A shackle on one leg sees that leg's tension, not the leg's share of the load. And the hook and master link carry the whole load.

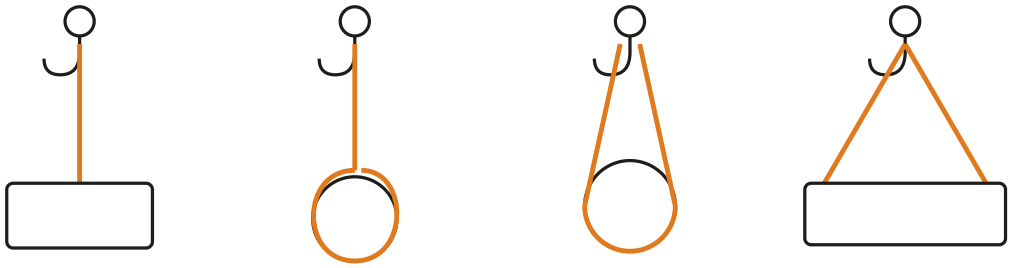
CHECK YOUR UNDERSTANDING

1. Two 2,000 kg synthetic slings in a bridle at 60°. What is the most they can lift?

Answers at the back of the chapter.

7 Chokers, baskets and multi-leg bridles

The hitch changes the sling's capacity before the angle is even considered.



The four hitches in this chapter: vertical, choker, basket and two-leg bridle.

Choker hitch

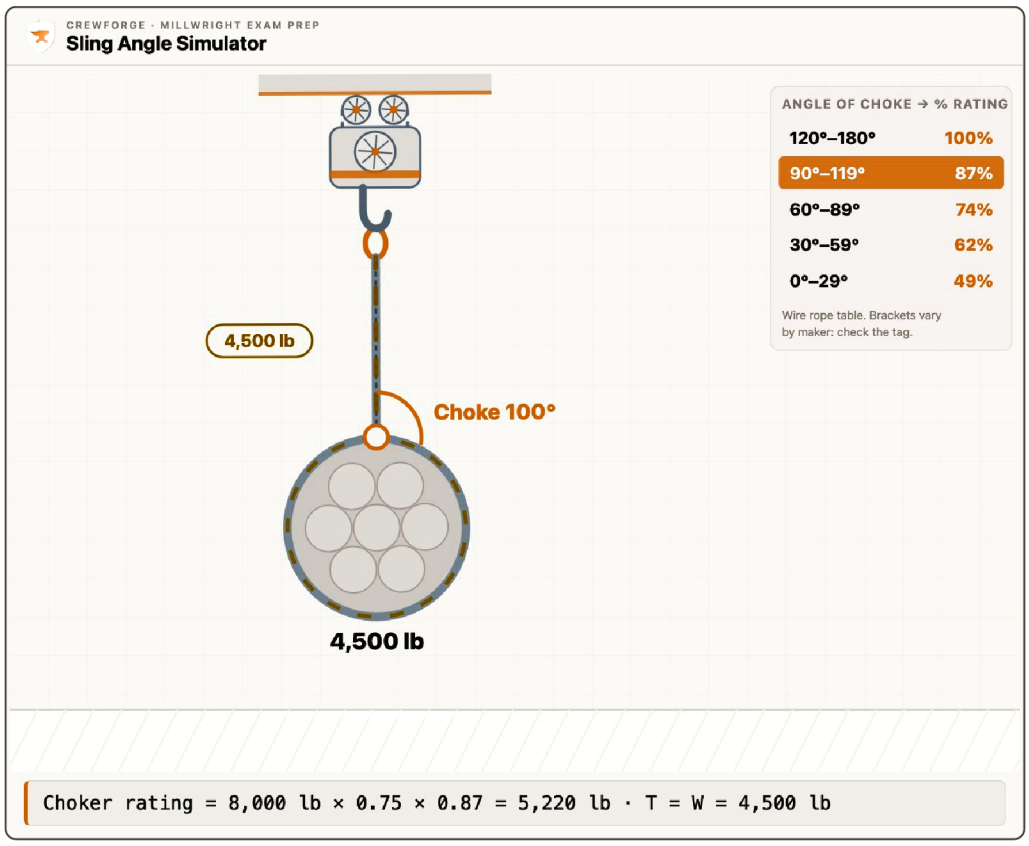
In a choker hitch the sling passes around the load and back through its own eye. The sling bends sharply where it passes through the eye, so a choker is rated lower than a vertical hitch: typically 75 % for wire rope, and 80 % for alloy chain and synthetic slings (Crosby catalogue), when the choke is 120° or more. The sling's tag or the maker's chart always governs.

If the choke is pulled down so the **angle of choke** is less than about 120°, the rating drops again. A choke left to settle on its own usually ends up near 135°, so no further reduction applies.

The table below is for wire rope slings. For a synthetic or chain choker pulled below 120°, use the sling maker's chart, because the figures differ by maker.

ANGLE OF CHOKE	% OF CHOKER RATING
120°–180°	100 %
90°–119°	87 %
60°–89°	74 %
30°–59°	62 %
0°–29°	49 %

Angle-of-choke reductions as published by Certex (Table 1) and Kennedy Wire Rope & Sling (Capacities and Hitches, PDF p. 1); the 120° edge is Crosby's (catalogue p. 471). Brackets vary slightly between makers.



Choke pulled to 100°: 8,000 × 0.75 × 0.87 = 5,220 lb rating.

EXAMPLE 7.1 **Choker with a pulled-down choke**

A wire rope sling rated 6,000 lb vertical is used as a choker. The choke is pulled down to 95°. What is its capacity? What would it be if the choke settled at 135°?

1. Choker rating: 6,000 × 0.75 = 4,500 lb
2. 95° is in the 90°–119° row: 87 %
3. 4,500 × 0.87 = 3,915 lb
4. At 135° (120°–180° row): 4,500 × 1.00 = 4,500 lb

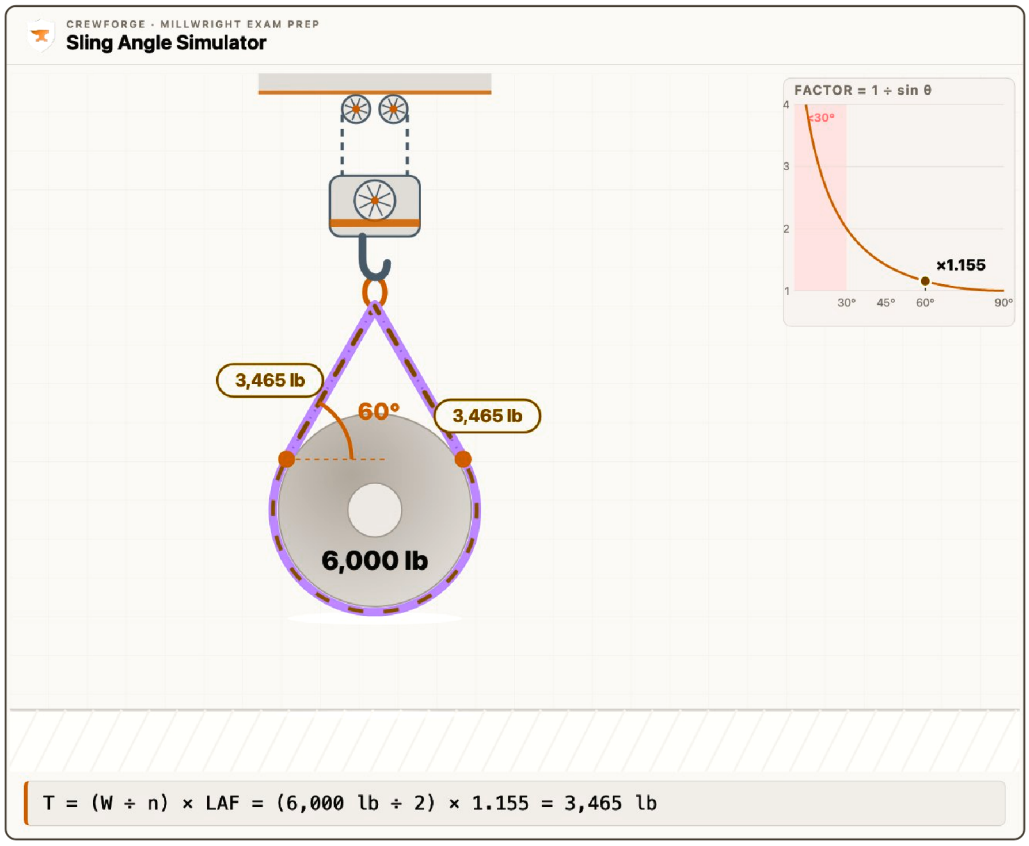
Answer: 3,915 lb at 95°; 4,500 lb at 135°

Basket hitch

In a basket the sling supports the load from underneath, with both eyes on the hook. With both legs vertical, two parts of sling share the load, so a basket is rated at twice the vertical rating.

Once the legs are angled, the load angle factor applies exactly as it does for a bridle. The load must also be large enough in diameter for the sling to bend around it without losing strength: check the D/d ratio on the maker's chart.

```
Basket, vertical legs: capacity = 2 × WLL
Basket, angled legs: capacity = 2 × WLL ÷ LAF
```



A basket around a round load with the legs at 60°: the factor applies to both legs.

EXAMPLE 7.2 **Angled basket**

A synthetic sling rated 4,000 lb vertical is used as a basket with the legs at 60°. What is its capacity?

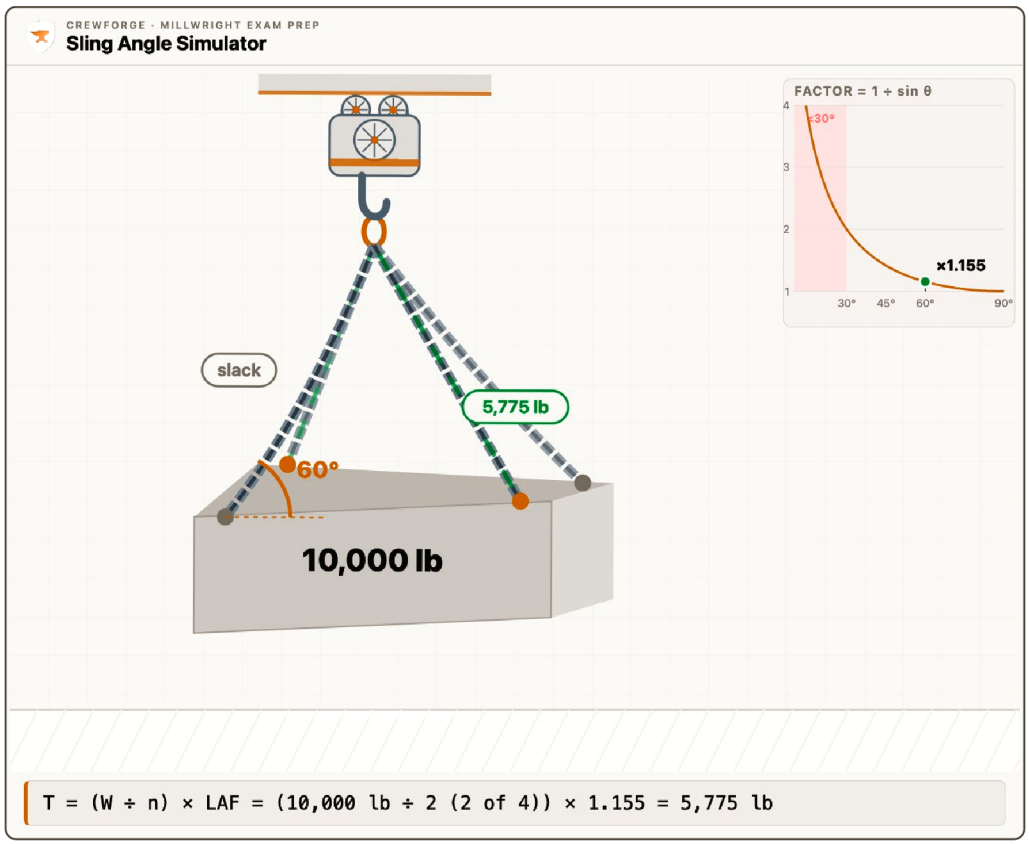
1. Vertical basket would be 2 × 4,000 = 8,000 lb
2. Angled: 8,000 ÷ 1.155 = 6,926 lb

Answer: 6,926 lb, not the 8,000 lb a vertical basket would allow

Three- and four-leg bridles: the two-leg rule

It seems a four-leg bridle should share the load four ways. On a rigid load it doesn't. The legs are never exactly the same length, the pick points are never exactly level, and the centre of gravity is never exactly centred. In practice the weight shifts onto two legs while the others carry little or nothing.

IHSA's rigging chapter warns that a four-leg load may be carried by only two legs while the others just balance it, and ITI's Toolbox Talk 42 says to assume only two legs make the lift. The safe practice is to plan a three- or four-leg bridle as if **only two legs carry the whole load**.



Four-leg bridle with the two-leg rule: 10,000 lb at 60° gives 5,775 lb per leg, double the 'paper' number.

EXAMPLE 7.3 **Four-leg bridle**

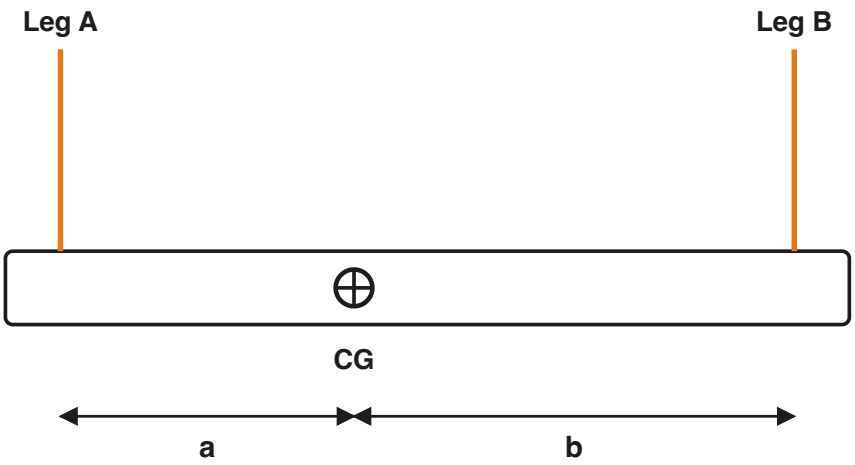
A 16,000 lb skid is lifted with a four-leg bridle at 50° from horizontal. Using the two-leg rule, what must each leg be rated for?

1. Two legs carry the load: 16,000 ÷ 2 = 8,000 lb
2. 50° → LAF 1.305
3. 8,000 × 1.305 = 10,440 lb

Answer: At least 10,440 lb per leg. Dividing by four would give half that, and overload the rig.

When the centre of gravity isn't centred

If the centre of gravity is closer to one pick point, that leg carries more. With vertical legs, each leg's share is found from the distances, the same way you'd balance a beam: the leg nearer the CG carries the larger share.



The leg nearer the CG carries more

Load shares with the centre of gravity off centre. Each leg carries the load times the distance to the other leg, divided by the distance between legs.

```
Leg A = W × b ÷ (a + b)
Leg B = W × a ÷ (a + b)
```

EXAMPLE 7.4 **Off-centre load, vertical legs**

A 6,000 lb load is lifted from two pick points 10 ft apart. The centre of gravity is 4 ft from pick point A. How much does each vertical leg carry?

1. a = 4 ft (CG to A), b = 6 ft (CG to B)
2. Leg A = 6,000 × 6 ÷ 10 = 3,600 lb
3. Leg B = 6,000 × 4 ÷ 10 = 2,400 lb
4. Check: 3,600 + 2,400 = 6,000 lb

Answer: Leg A 3,600 lb, leg B 2,400 lb. Size both slings for the heavier leg.

ON THE JOB

Find the centre of gravity first

With angled legs and an off-centre load the split is more complex, and the leg nearer the CG still carries more. Work out where the CG is before you pick, and get the hook directly above it. If the load tilts when it leaves the ground, set it down and re-rig.

CHECK YOUR UNDERSTANDING

1. A wire rope sling rated 2,000 kg is used as a choker with the choke settled at 140°. What is its capacity?
2. A four-leg bridle lifts 8,000 kg at 45°. Using the two-leg rule, what is the tension in each leg?

Answers at the back of the chapter.

8 Putting it together: planning a two-leg lift

On the job, sling angles are one step in a lift plan, not a math problem on its own.

Here is the order a millwright works through for a routine two-leg lift. Your site's lift plan and procedures come first; this is the thinking behind them.

1. Know the weight: nameplate, drawings or the maker's data. Never guess.
2. Find the centre of gravity and choose pick points on either side of it.
3. Check the pick points: lugs and eye bolts rated for the load and for the angle of pull.
4. Choose the hitch, then work out the sling angle from L and H (or L and S).
5. Calculate the tension in each leg with the load angle factor.
6. Choose slings and hardware with a WLL at or above that tension for the hitch used.
7. Check headroom: hook height, sling length and the height the load must clear.
8. Confirm the plan with your lift supervisor and signal person, and lift a few inches first to check balance.

EXAMPLE 8.1 **Field scenario: replacing a gearbox (metric)**

You're replacing a 4,600 kg gearbox. The maker's drawing shows its two lifting lugs 1.8 m apart, centred on the CG. The crib has synthetic slings rated 3,000 kg in 2.0 m and 1.5 m lengths. The lift drawing gives the hook room up to 2.0 m above the lugs. Which slings do you use?

1. 2.0 m slings: H = √(2.0² − 0.9²) = 1.786 m, which fits under 2.0 m
2. LAF = 2.0 ÷ 1.786 = 1.120 (about 63°)
3. Tension: (4,600 ÷ 2) × 1.120 = 2,576 kg, 86 % of the 3,000 kg rating
4. 1.5 m slings: H = √(1.5² − 0.9²) = 1.200 m; LAF = 1.5 ÷ 1.200 = 1.250 (about 53°)
5. Tension: 2,300 × 1.250 = 2,875 kg, 96 % of the rating

Answer: Use the 2.0 m slings: 2,576 kg per leg. The 1.5 m slings stay within rating but leave little margin (96 %), and save headroom you don't need.

SAFETY

Exam preparation, not a procedure

This chapter teaches the calculations behind a lift. On the job, follow the sling tag, the manufacturer's data, your site's procedures and the lift plan. Never stand under a load.

Key formulas

Load angle factor	`LAF = 1 ÷ sin θ = L ÷ H`
Tension per leg	`T = (W ÷ n) × LAF`
Angle conversions	`θ = 90° − from vertical; θ = 90° − included ÷ 2`
Height from pick-point distance	$H = \sqrt{(L^2 - (S \div 2)^2)}$
Most a two-leg rig can lift	`2 × WLL ÷ LAF`
Choker	`WLL × 75 % (wire rope) × angle-of-choke %; synthetic 80 %, maker's chart below 120°`
Basket	`2 × WLL vertical; 2 × WLL ÷ LAF angled`
Three- and four-leg bridles	`n = 2 (two-leg rule)`
Off-centre CG, vertical legs	`Leg A = W × b ÷ (a + b)`

Where people lose marks

- **Using the angle from vertical, or the included angle, as the sling angle.** The formula needs the angle from horizontal. Convert first: 90° minus the angle from vertical, or 90° minus half the included angle.
- **Dividing by four on a four-leg bridle.** On a rigid load, two legs end up carrying the weight. Plan for two.
- **Using H ÷ L instead of L ÷ H.** The factor is never less than 1. If it is, flip it.
- **Forgetting the choker rating, or the angle-of-choke reduction.** A choker starts at 75 % (wire rope) or 80 % (synthetic) of vertical, and drops again below a 120° choke: by the table for wire rope, by the maker's chart for synthetic.
- **Claiming the full 200 % basket rating when the basket legs are angled.** Twice the vertical rating applies only with vertical legs. Divide by the load angle factor once they're angled.
- **Calculator in radians.** sin 30 must give 0.5. If it doesn't, switch to degrees.
- **Checking the slings but not the hardware.** Shackles, lugs and eye bolts see the leg tension, often at an angle. Check them too.

Practice questions

Part A is multiple choice, in the style of the Red Seal exam. Part B asks you to show your working. Answers and full solutions are at the back.

Part A

A1. A 12,000 lb load is lifted with a two-leg bridle at 60° from horizontal. What is the tension in each leg?

A. 6,000 lb
B. 6,930 lb
C. 8,484 lb
D. 12,000 lb

A2. A 3,000 kg load hangs from a two-leg bridle with a 120° included angle between the legs. What is the tension in each leg?

A. 1,500 kg
B. 1,733 kg
C. 3,000 kg
D. 6,000 kg

A3. Two 10 ft slings lift a 6,000 lb load. The hook is 8 ft above the pick points. What is the tension in each leg?

A. 2,400 lb
B. 3,000 lb
C. 3,750 lb
D. 7,500 lb

A4. A wire rope sling rated 4,000 lb vertical is used in a choker hitch with a 75° angle of choke. What is its adjusted capacity?

A. 4,000 lb
B. 3,000 lb
C. 2,960 lb
D. 2,220 lb

A5. A 20,000 lb load is lifted with a four-leg bridle at 60° from horizontal. Using the two-leg rule, what tension must each leg be rated for?

A. 5,775 lb
B. 10,000 lb
C. 11,550 lb
D. 23,100 lb

A6. A 2,400 kg pump is lifted with a two-leg bridle at 45° from horizontal. What is the minimum rating required for each leg?

A. 1,200 kg
B. 1,697 kg
C. 2,400 kg
D. 3,394 kg

A7. A 4,000 lb load is lifted with a two-leg bridle. Each leg is 30° from vertical. What is the tension in each leg?

A. 2,000 lb
B. 2,310 lb
C. 2,828 lb
D. 4,000 lb

Part B

B1. A 14,000 lb load is lifted on two legs at 55° from horizontal. Find the tension in each leg.

B2. A 2,500 kg load hangs on two legs, each 40° from vertical. Find the tension in each leg.

B3. Two 10 ft slings lift a 6,000 lb load. The hook is 8 ft above the pick points. Find the tension in each leg.

B4. A wire rope sling rated 5,000 kg vertical is used as a choker with a 70° angle of choke. What is its capacity?

B5. A four-leg bridle lifts a 20,000 kg module with legs at 60°. Using the two-leg rule, what must each leg be rated for?

B6. A sling rated 6,000 lb vertical is used as a basket with the legs at 45°. What is its capacity?

B7. Two 3,000 lb slings are rigged at 30° to lift a 5,500 lb load. Is the lift within rating?

B8. A 9,000 lb load has its centre of gravity 3 ft from pick point A and 6 ft from pick point B. With vertical legs, how much does each leg carry?

Answers and worked solutions

Check your understanding

Section 2.

1. 2,000 lb. With vertical legs the factor is 1.000, so each leg carries exactly its share.
2. Nothing. The vertical part stays W ÷ n. Only the horizontal part and the tension grow.

Section 3.

1. 1 ÷ sin 70° = 1.064
2. 2,500 × 1.155 = 2,888 lb
3. 900 × 2.000 = 1,800 kg

Section 4.

1. 90° – 40° = 50°
2. 90° – 60° = 30°; LAF = 2.000

Section 5.

1. 8 ÷ 6.4 = 1.250
2. √(16 – 1) = √15 = 3.87 m

Section 6.

1. 2 × 2,000 ÷ 1.155 = 3,463 kg

Section 7.

1. 2,000 × 0.75 = 1,500 kg (no angle-of-choke reduction above 120°)
2. (8,000 ÷ 2) × 1.414 = 5,656 kg

Part A

A1. B. 6,930 lb

(12,000 ÷ 2) × 1.155 = 6,930 lb

A2. C. 3,000 kg

120° included → 30° from horizontal → (3,000 ÷ 2) × 2.000 = 3,000 kg

A3. C. 3,750 lb

L ÷ H = 10 ÷ 8 = 1.250 → (6,000 ÷ 2) × 1.250 = 3,750 lb

A4. D. 2,220 lb

4,000 × 0.75 (choker) × 0.74 (60°–89° choke) = 2,220 lb

A5. C. 11,550 lb

Two legs carry: (20,000 ÷ 2) × 1.155 = 11,550 lb

A6. B. 1,697 kg

(2,400 ÷ 2) × 1.414 = 1,697 kg

A7. B. 2,310 lb

30° from vertical → 60° from horizontal → (4,000 ÷ 2) × 1.155 = 2,310 lb

Part B

B1. 8,547 lb

1. LAF = 1 ÷ sin 55° = 1.221
2. (14,000 ÷ 2) × 1.221 = 8,547 lb

B2. 1,631 kg

1. θ = 90° − 40° = 50°
2. LAF 1.305
3. 1,250 × 1.305 = 1,631 kg

B3. 3,750 lb

1. LAF = 10 ÷ 8 = 1.250
2. 3,000 × 1.250 = 3,750 lb

B4. 2,775 kg

1. Choker: 5,000 × 0.75 = 3,750 kg
2. 70° → 74 %
3. 3,750 × 0.74 = 2,775 kg

B5. 11,550 kg (11.55 t)

1. Two legs: 20,000 ÷ 2 = 10,000 kg
2. 10,000 × 1.155 = 11,550 kg

B6. 8,487 lb

1. 2 × 6,000 ÷ 1.414 = 8,487 lb

B7. No. Each leg would carry 5,500 lb on a 3,000 lb sling, and 30° is already the lowest angle allowed without an engineered plan.

1. Most the rig can lift: 2 × 3,000 ÷ 2.000 = 3,000 lb
2. 5,500 lb is far above 3,000 lb
3. Each leg would see 2,750 × 2.000 = 5,500 lb

B8. Leg A 6,000 lb, leg B 3,000 lb

1. Leg A = 9,000 × 6 ÷ 9 = 6,000 lb
2. Leg B = 9,000 × 3 ÷ 9 = 3,000 lb

Sources

- Red Seal Occupational Standard (RSOS), Industrial Mechanic (Millwright), Major Work Activity B: Task B-9 Plans rigging, hoisting/lifting and moving; Task B-10 Rigs, hoists/lifts and moves load (red-seal.ca/eng/trades/industrialmech/mwa-b.shtml)
- IHSA (Ontario), Construction Health and Safety Manual, Chapter 24 Rigging: bridle hitches, hook directly over the centre of gravity, a four-leg load may be carried by only two legs (ihsa.ca/rtf/health_safety_manual/pdfs/equipment/Rigging.pdf)
- ASME B30.9, Slings, by topic only (clause text not reproduced), as summarised by Mazzella (design factors: chain 4 : 1, wire rope and synthetic 5 : 1; mazzellacompanies.com/?p=1871) and OSHA Guidance on Safe Sling Use (no horizontal angles under 30° except as the maker or a qualified person recommends; tags give the rated load for each hitch; osha.gov/safe-sling-use)
- The Crosby Group catalogue (2019), p. 471: wire rope choker 75 % and chain choker 80 % of a single leg when the choke is 120° or more; basket twice a single leg only with vertical legs and D/d 25 : 1; Crosby catalogue 2023, p. 416: synthetic choker 80 % (thecrosbygroup.com)
- Certex, Table 1, choker hitch rated capacity adjustment (100, 87, 74, 62, 49 %) and the normal choke of about 135° (certex.com/?p=2537); Kennedy Wire Rope & Sling, Capacities and Hitches (PDF p. 1)
- Trinity Sling, wire rope slings: bridles tagged for vertical, 60°, 45° and 30° (trinitysling.com/wire-rope-slings)
- OSHA, Guidance on Safe Sling Use, wire rope, alloy chain and synthetic web pages: daily inspection before use and removal-from-service criteria (osha.gov/safe-sling-use)
- Industrial Training International, Toolbox Talk 42, Sometimes 4 = 2: with a 4-leg bridle, assume only 2 legs make the lift (iti.com/hubfs/ToolboxTalks_42_041015.pdf)
- Load angle factors computed as 1 ÷ sin θ and rounded to three decimals; every worked example checked against the CrewForge sling angle simulator's calc()

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